

Niels Bohr Quantum Summer School Challenge

Quantum simulation of non-linear PDEs

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Abstract

Solving nonlinear partial differential equations is one of the main objectives in scientific computing and account for a substantial share of modern supercomputing workloads. Although quantum algorithms can offer significant advantages for linear problems, the simulation of non-linear systems remains a major challenge. In this challenge, you will investigate quantum algorithms for solving either the viscous Burgers equation $\partial_t u + u\partial_x u = \nu\partial_{xx} u$ or the Korteweg-de Vries equation $\partial_t u + u\partial_x u + \delta\partial_{xxx} u = 0$.

1 Background

There are several quantum algorithms that can effectively solve linear differential equations [4, 6, 3, 11, 5]

$$\dot{x}(t) = Ax(t) \quad \text{subject to} \quad x(0) = x_0, \quad (1)$$

for $x(t) \in \mathbb{C}^N$ and $A \in M_N(\mathbb{C})$ in the presence of appropriate assumptions on the system. The Hamiltonian simulation problem is the special case $A = iH$ with H Hermitian. Due to the linear nature of quantum computers, more general non-linear equations of the form

$$\dot{x}(t) = f(t, x(t)) \quad \text{subject to} \quad x(0) = x_0, \quad (2)$$

are more challenging. Nevertheless, for several families of polynomial vector fields f , there are efficient quantum solutions ([16, 13, 10, 2, 9, 15]).

Partial differential equations (PDEs) introduce another layer of complexity. Linear PDEs can in many cases be handled effectively on a quantum computer [7]. Examples include Poisson's equation $\Delta\phi = f$, the heat equation $\partial_t\phi = \Delta\phi$ [14] and the wave equation $\partial_t^2\phi - \Delta\phi = 0$ [8]. Efficient quantum algorithms for non-linear PDEs are more elusive and there have been established limitations in this direction [1]. However, the numerical solution of nonlinear PDEs, such as the Navier-Stokes equations, remains one of the central challenges in scientific computing, and advances in this area could have far-reaching societal impact.

2 The Challenge

Develop a quantum algorithm for solving one of the following basic non-linear PDEs.

- (1) The viscous Burger's equation with viscosity coefficient $\nu > 0$

$$\partial_t u + u\partial_x u = \nu\partial_{xx} u \quad (3)$$

(2) The Kortevæg-de Vries (KdV) equation with dispersion coefficient $\delta > 0$.

$$\partial_t u + u \partial_x u + \delta \partial_x^3 u = 0 \quad (4)$$

In both cases, $u = u(t, x) : [0, \infty) \times [0, 1] \rightarrow \mathbb{R}$ subject to initial conditions $u(0, x) = u_0(x)$ and periodic boundary conditions $u(t, 0) = u(t, 1)$ for all t . The initial function is taken to be smooth and periodic. This ensures that the solution exists and is unique.

- You can assume access to an oracle preparing an amplitude encoding of the initial function

$$|0\rangle \mapsto \frac{1}{\beta} \sum_{j=0}^{2^n-1} u_0(x_j) |j\rangle \quad (5)$$

for a normalizing constant β and grid points $x_j \in [0, 1]$.

- Most approaches will require you to first linearise the equation before you start to design a quantum algorithm. Known methods for this include Carleman linearisation [10, 16], Koopman von Neumann linearisation [12] and the homotopy perturbation method [17].
- The goal is to prepare an amplitude encoding (as in Eq. (5)) of a numerical approximation of the solution at time $T > 0$.
- Estimate the resources required by your algorithm (query or gate complexity, qubit counts, success probability if applicable).
- If feasible, implement a small instance of your solution in Qiskit. Suggested initial conditions are $u_0(x) = \sin(2\pi x)$ for Burgers and $u_0(x) = \cos(4\pi x)$ or $u_0(x) = \text{sech}(40(x-1/2))^2$ for KdV.

3 Some useful bounds

Given the assumptions presented in the previous section, one can establish the following bounds for Burgers equation.

$$\|u(t)\|_2 \leq \|u_0\|_2 \quad \text{and} \quad \|u(t)\|_\infty \leq \|u_0\|_\infty. \quad (6)$$

Similarly, for the KdV equation one can establish the following.

$$\|u(t)\|_2 = \|u_0\|_2 \quad \text{and} \quad \|u(t)\|_\infty \leq C(\|u_0\|_2^2 + \|\partial_x u_0\|_{L^2}^2)^{1/2}, \quad (7)$$

for some constant C . The point is that both the L^2 norm and the L^∞ norms are bounded by a quantity determined by the initial condition u_0 . This is typically a necessary condition in order to enable efficient quantum simulation.

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